

Introduction to Microeconomics

Lecture 10—Consumer Theory: The Consumer's Problem

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

ECON101
Fall 2023

The Consumer's Problem

The Consumer's Problem

Choose the best bundle, according to the consumer's preferences subject to the consumer's budget constraint. → Choose the best among what is feasible.

$$\begin{aligned} \max_{(x_1, x_2, \dots, x_n) \in \mathbb{R}_+^n} & \quad u(x_1, x_2, \dots, x_n) \\ \text{subject to:} & \quad p_1 x_1 + p_2 x_2 + \dots + p_n x_n \leq I \end{aligned}$$

where $p_1, p_2, \dots, p_n$ are the prices of the n goods and I is income.

Two-good world

$$\begin{aligned} \max_{(x_1, x_2) \in \mathbb{R}_+^2} & \quad u(x_1, x_2) \\ \text{subject to:} & \quad p_1 x_1 + p_2 x_2 \leq I \end{aligned}$$

$$\max_{(x_1, x_2) \in \mathbb{R}_+^n} u(x_1, x_2) \quad \text{subject to: } p_1 x_1 + p_2 x_2 \leq I$$

Suppose that preferences are:

- ① Complete
- ② Transitive
- ③ Monotonic (More is better)
- ④ Strictly Convex (strict preference for balancedness) $\leftrightarrow$ (MRS strictly decreasing)

The optimality condition

We can simplify the problem to: $\max_{(x_1, x_2) \in \mathbb{R}_+^n} u(x_1, x_2)$ subject to: $p_1 x_1 + p_2 x_2 = I$
Furthermore, the consumer's optimal consumption bundle satisfies:

$$MRS = \frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

Cobb-Douglas Utility: $u(x_1, x_2) = x_1 x_2$

Perfect Complements

Suppose that I like to prepare my sandwiches with exactly 2 slices of Bread and 1 slice of Ham.

Special Cases: Corner Solutions

Perfect Complements: $u(x_1, x_2) = 2x_1 + x_2$